

Module 2

1. Brief about SplitMAC architecture and how it improves the AP's performance

SplitMAC means the MAC (Medium Access Control) functions of Wi-Fi are divided between the **Access Point (AP)** and the **Wireless LAN Controller (WLC)**.

Time-critical tasks like sending beacons, acknowledgments, and encryption are handled by the AP, while management tasks like authentication, roaming, and configuration are handled by the controller.

This reduces the processing load on the AP and allows centralized management from the WLC.

It improves scalability, easier maintenance, fast roaming, and better RF optimization in large networks.

This architecture is mainly used in **Lightweight APs (LWAPs)** with CAPWAP.

2. Describe CAPWAP and explain the flow between AP and Controller

CAPWAP (Control And Provisioning of Wireless Access Points) is a protocol used for communication between a Lightweight AP and a Wireless LAN Controller.

When the AP boots, it gets an IP address using DHCP and discovers the WLC using DHCP option, DNS, or broadcast.

The AP then forms a secure CAPWAP tunnel with the controller and downloads configuration/software if needed.

After joining, all control information like authentication and AP management flows through the control tunnel.

User traffic may also pass through the WLC depending on the AP mode.

3. Where does CAPWAP fit in the OSI model, and what are the two tunnels in CAPWAP?

CAPWAP mainly works at the **Application Layer (Layer 7)** of the OSI model, but it uses **UDP** at the transport layer.

It uses **UDP port 5246** for control traffic and **UDP port 5247** for data traffic.

CAPWAP creates two logical tunnels between AP and controller:

1. **Control Tunnel** – carries management traffic like AP registration, configuration, authentication, keepalives, and RF management.
2. **Data Tunnel** – carries actual wireless client data packets between AP and WLC.

The control tunnel is always encrypted using DTLS for security.

4. Difference between Lightweight APs and Cloud-based APs

A **Lightweight AP** depends on a physical or virtual Wireless LAN Controller inside the organization network.

The WLC manages configurations, roaming, security, and RF optimization centrally.

A **Cloud-based AP** connects to a cloud controller over the internet instead of a local WLC.

Cloud APs are easier to deploy and manage remotely from anywhere using a web dashboard.

Lightweight APs are common in enterprise campuses, while cloud APs are popular in distributed branches and small businesses.

5. How is the CAPWAP tunnel maintained between AP and Controller?

After the AP joins the WLC, the CAPWAP tunnel stays active using **keepalive messages** called heartbeats or Echo Requests.

The AP and controller continuously exchange these packets to confirm both devices are alive.

If the AP does not receive responses for a certain time, it assumes the controller is unreachable.

The AP then tries to rediscover and reconnect to the same or backup controller.

This mechanism ensures reliable communication and automatic recovery.

6. Difference between Sniffer mode and Monitor mode, with use case

Sniffer Mode is used to capture wireless packets for troubleshooting and packet analysis.

The AP sends captured packets to tools like Wireshark for debugging issues like roaming, authentication, or packet loss.

It is mainly used by network engineers for deep packet inspection.

Monitor Mode is used for security and RF monitoring.

The AP does not serve clients; instead, it scans channels looking for rogue APs, interference, or attacks.

It is mainly used for wireless intrusion detection and RF optimization.

7. If WLC is deployed in WAN, which AP mode is best for local network and how?

When the WLC is located across a WAN, **FlexConnect mode** is the best option.

In FlexConnect, the AP locally switches client traffic instead of sending all traffic to the remote controller.

Only management and control traffic go through the CAPWAP tunnel to the WLC.

This reduces WAN bandwidth usage and improves performance for branch users.

Even if the WAN link fails, local users may still access local network resources.

8. Challenges of deploying autonomous APs in large networks like universities

Autonomous APs are configured individually, so managing more than 50 APs becomes difficult and time-consuming.

There is no centralized controller for configuration, firmware updates, or security policies.

Roaming between APs is slower because APs do not coordinate with each other efficiently.

RF channel and power optimization must be done manually, leading to interference issues.

Troubleshooting and monitoring also become harder in large environments.

9. What happens when a wireless client connected to a Lightweight AP loses connection to the WLC?

In Local Mode, the Lightweight AP depends heavily on the WLC.

If the WLC goes down, the AP loses its CAPWAP connection and cannot process new client authentications.

Existing clients may continue working for some time depending on the vendor and configuration, but eventually connectivity is affected.

New users usually cannot join the network until the controller is restored.

To avoid this issue, enterprises use redundant WLCs or FlexConnect mode for branch deployments.